Preserving ATPD-2352 Revision U Durability in Operational Service

Embedded laminate temperature sensing for heated transparent armor windshields

A summary of the HLG white paper on how embedded laminate temperature sensing and closed-loop heater control can help heated transparent armor windshields stay within the durability envelope validated during ATPD-2352 Revision U qualification.

1. The issue

ATPD-2352 Revision U expands environmental conditioning and accelerated aging requirements for transparent armor used on military ground vehicles. Qualification testing cannot reproduce every service condition, and one important operational variable is the heat generated by electrically heated windshields.

Many fielded heater systems use open-loop control: power is applied without measuring laminate temperature. Over long de-icing periods this can add cumulative thermal dwell exposure to the laminate.

2. Why time at temperature matters

Polymer interlayers are viscoelastic. As temperature rises, stiffness falls and creep under sustained stress increases. Sustained exposure may contribute to:

- interlayer creep and stress redistribution
- progressive weakening of adhesive interfaces
- moisture migration along laminate edges
- micro-voids that can develop into delamination, haze or optical distortion

3. Why windshields are most exposed

Windshields usually carry the heated interlayer, are the largest exposed transparent armor surface, and are tied directly to driver visibility. They therefore tend to see the highest thermal duty cycle of any armored glazing component.

4. Established practice in other sectors

Embedded laminate temperature sensing with closed-loop heater control is long-established in heated transparencies for military and commercial aircraft, naval bridge glazing, marine heated windows and rail glazing. Applying it to transparent armor is an adaptation of existing practice, not a new or experimental technology.

5. Open-loop vs closed-loop control

Open-loop systems regulate electrical power only. Closed-loop systems use embedded sensors to reduce or pause heater output as laminate temperature approaches defined limits, while still providing de-icing. Benefits include:

- active control of laminate temperature
- reduced cumulative thermal dwell exposure
- more predictable lifecycle behavior

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6. Integration approach

Sensors are typically placed in a dedicated sensing layer, separate from the heating interlayer, and routed to the heater controller through the windshield harness. The approach fits within normal laminate manufacturing and does not require a fundamental change to the transparent armor construction. It could also be introduced at routine windshield replacement.

7. Conclusion

Embedded laminate temperature sensing helps keep operational heater use aligned with the conditions assumed during Revision U qualification. It should be viewed as a practical safeguard for the Rev U investment made by government and industry, helping qualification gains carry into service life.

The full 16-page HLG white paper (WP-001) covers each topic in more depth, including an appendix on the operational thermal envelope. It is available to customers on request. Request it through the Engineering Review form at www.hotlineglassusa.com or by email to Hgil@Hotlineglassusa.com.

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